

# Samir Varma

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## EDUCATION

### Rutgers University, New Brunswick - Honors College

GPA: 3.936/4.000

*B.S. in Computer Engineering, B.S. in Mathematics, Minor in Physics*

*Aug. 2023 – May 2027*

- Relevant Coursework: Distributed Deep Learning, Software Engineering, Machine Learning, Computer Systems

## EXPERIENCE

### Software Engineering Intern

May 2026 – August 2026

*Dropbox*

*San Francisco, CA*

- Built an automated Quarterly Business Review deck generation workflow using **Python, Flask, LangGraph, Redis, and LangChain**, managing service builds with **Bazel** and integrating live customer data and LLM narratives.
- Engineered a cloud data-grounding layer using **Databricks SQL and SQLAlchemy** with 11 parameterized query sources for trailing-12-month YoY analytics, cutting QBR drafting time by **95%**.
- Architected a conversational editing pipeline with parallel data grounding, structured LLM outputs, schema-validated block mutations, optimistic concurrency control, and audit logging, achieving a **92%** task success rate.

### Data Science Intern

June 2025 – August 2025

*AT&T*

*Middletown, NJ*

- Built and deployed **Isolation Forest-based intrusion detection services** via **Scikit-Learn** to flag malicious IPs from high-volume network telemetry, processing **3M+ traffic records** with low-latency inference and increasing anomaly detection precision to **96%** through **GridSearchCV** parameter sweeps.
- Engineered a scheduled ETL workflow using **Apache Airflow** to orchestrate data ingestion, validation, and feature preprocessing from **Azure Databricks**, enabling reliable daily model refreshes.

### Software Engineering Intern

January 2025 – April 2025

*Cocogen.ai*

*San Francisco, CA*

- Developed AI inference services using **Python, FastAPI, PyTorch, Docker, and Google Cloud Platform**, orchestrating computer vision workflows to automated user-driven image editing via scalable backend services.
- Engineered distributed image processing pipelines using **Python, OpenCV, Redis, and GCS** to batch-process **5,000+** images, improving throughput and enabling reliable large-scale computer vision inference.

### Machine Learning Research Assistant

June 2024 – April 2025

*Rutgers University*

*Piscataway, NJ*

- Implemented **Actor-Critic reinforcement learning agents** in **PyTorch** within the CARLA autonomous driving simulator, designing modular policy, value, improving policy learning by **14%**.
- Built an experimental evaluation framework to compare **DQN** vs. **Actor-Critic architectures**, instrumenting reward curves, convergence behavior, and policy stability across parallel simulation runs.

## PROJECTS

### Blacksmith | *Python, Pandas, NumPy, Scikit-learn, Typer, Pydantic, Instructor, Chart.js*

- Engineered an **agentic data-cleaning CLI** that profiles messy CSV and Kaggle datasets, detects missing values, duplicates, outliers, and type inconsistencies, and exports cleaned data with quality scores and audit reports.
- Designed a structured LLM decision-making pipeline through **OpenAI, Anthropic, and xAI** providers, using schema validation, cached agent decisions, and a **Chart.js** dashboard for transparent cleaning analysis.

### PokéChat | *Next.js, Tailwind CSS, FastAPI, OpenAI API, Chart.js, Beautiful Soup*

- Developed a **full-stack competitive Pokémon team analysis website** using **FastAPI** backend with AI-powered team evaluation and **Next.js** frontend, implementing web scraping for real-time strategic data from Pikalytics.com
- Built comprehensive team validation and meta trend visualization tools with **Chart.js** and the **OpenAI API**, enabling players to analyze team compositions, strengths and weaknesses, and competitive trends.

### NextLevel | *Next.js, TypeScript, MongoDB, AWS S3, AWS Lambda*

- Designed a **full-stack game review platform** with **Next.js, TypeScript, and MongoDB**, using the IGDB API for **400K+** games and building secure authentication, REST APIs, and scalable data models.
- Implemented review creation and engagement features with a responsive **Tailwind CSS UI** and **AWS (S3, Lambda)** backend services, supporting **30+** active users with reliable media storage and low-latency page loads.

## SKILLS

**Languages:** Python, Go, C/C++, SQL, Java, JavaScript, TypeScript, HTML/CSS

**Tools:** AWS, GCP, Docker, Kubernetes, PyTorch, TensorFlow, React.js, Node.js, FastAPI, Flask, LangChain, OpenAI/Anthropic APIs, Pydantic, PostgreSQL, MySQL, MongoDB, Git/GitHub